

EDES 301

Adaptive Irrigation Controller Proposal

2/22/2026
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Background Information

An embedded closed-loop soil moisture control system that:

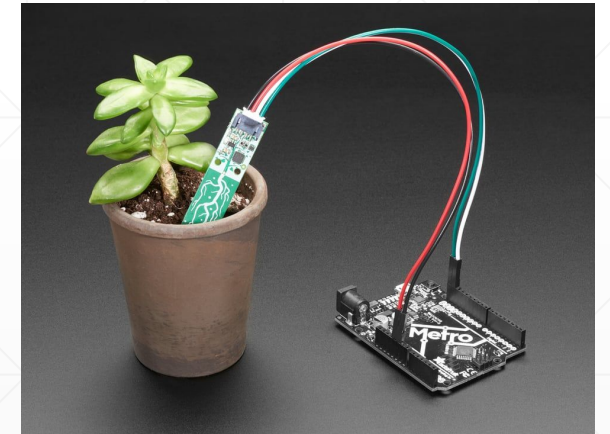
- Automatically irrigates a plant
- Monitors soil moisture continuously
- Models soil absorption and drying behavior
- Predicts future moisture trends
- Monitors reservoir level
- Detects system faults
- Provides visual system-state feedback via LEDs
- Boots autonomously on power-up

Most DIY irrigation systems:

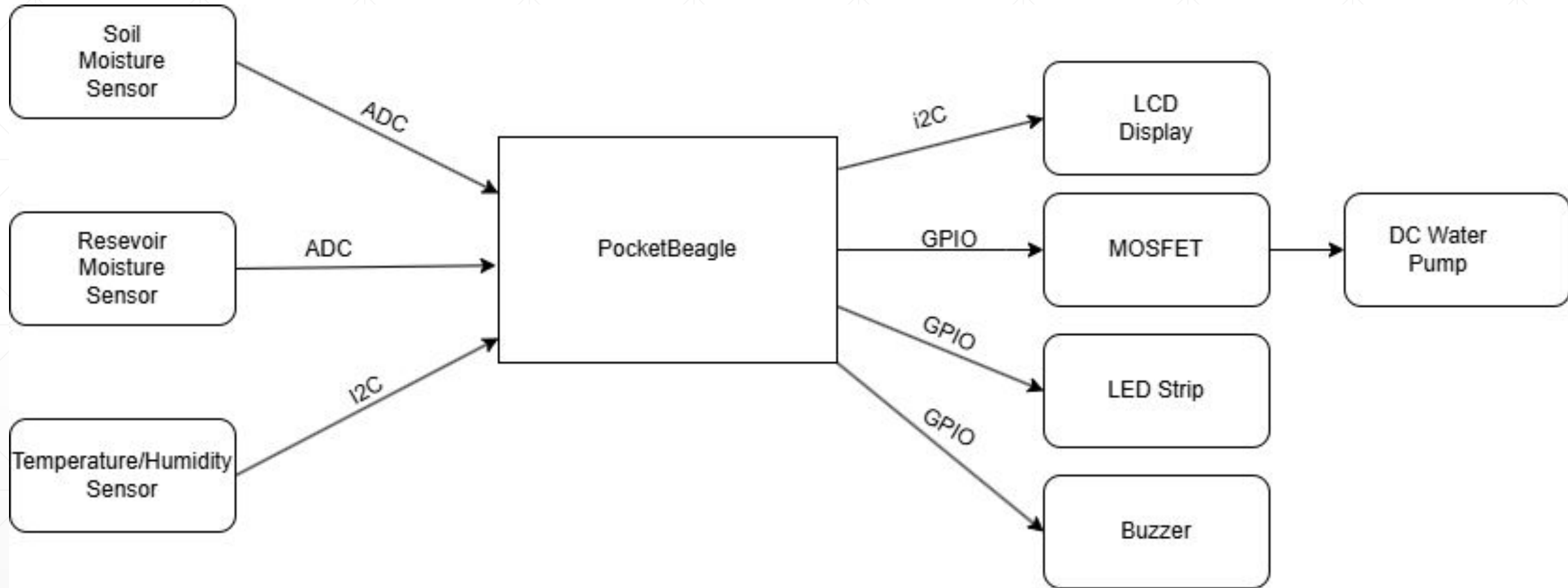
- Use simple threshold control
- Do not model soil behavior
- Do not detect faults
- Do not analyze moisture decay trends
(https://projecthub.arduino.cc/lc_lab/automatic-watering-system-for-my-plants-e4c4b9)

This project introduces:

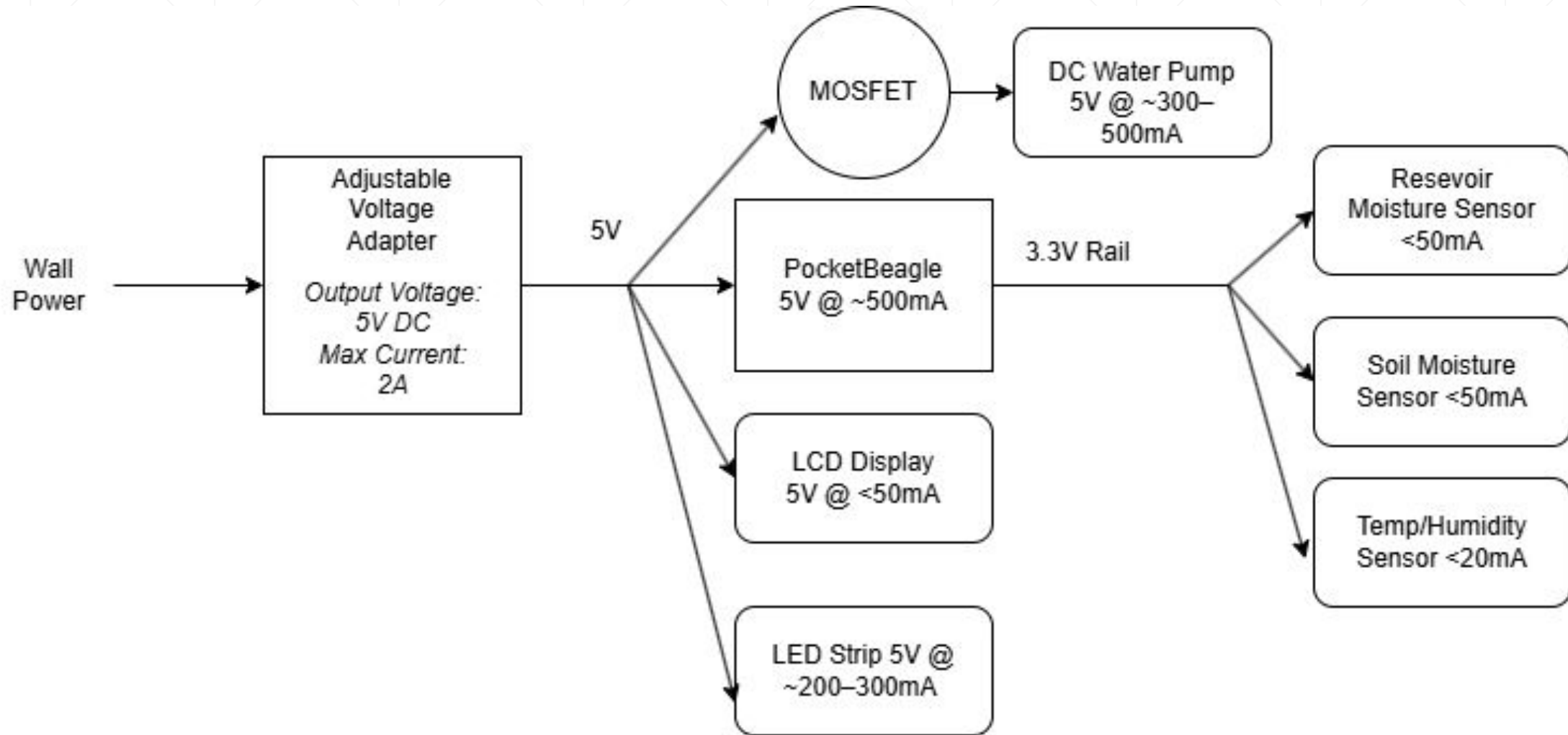
- Closed-loop control
- Soil absorption characterization
- Predictive watering logic
- Reservoir usage analytics
- Fault detection



System Block Diagram



Power Block Diagram



Components / Budget

Component	EDES301 to Buy?	Cost
<u>5 V Adjustable Voltage Adapter (2 A)</u> (on parts list)	Yes	15.98
<u>LCD Display 16x2</u> (on parts list)	Yes	9.95
<u>Moisture Sensor 1</u> (on parts list)	Yes	2.00
<u>Moisture Sensor 2</u> (on parts list)	Yes	2.00
<u>Temp / Humidity Sensor</u> (on parts list)	Yes	1.70
<u>DC Water Pump</u> (NOT on parts list)	No	6.99
<u>LED Strip</u> (NOT on parts list)	No	8.99

COST for EDES 301: \$31.63

TOTAL COST: \$47.61